

The empirical recurrence for OEIS A230929, proved by transfer matrix

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Abstract

OEIS A230929 counts the black-square subarrays of arrays of width 4 over $\{0, 1, 2, 3\}$ in which every cell has a diagonal neighbour one greater than it modulo 4, and none equal to it. The entry carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A diagonal step changes both coordinates by one and so never changes the colour of a cell, which makes the two colours of the chessboard independent of one another; the count is then a walk over the cells of one colour, on $S = 157$ states.

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1 The conjecture

OEIS A230929 is “Number of black-square subarrays of $(n+2) \times (2+2)$ 0..3 arrays $x(i,j)$ with each element diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$, no adjacent elements equal, and upper left element zero.”. It has offset 1 and begins

$$2, 6, 16, 48, 146, 438, 1312, 3936, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = 3 \cdot a(n-1) - a(n-2) + 3 \cdot a(n-3).$$

As of the “Last modified” line on the live entry (Mar 17 2018 07:10:17, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Colour the cells of the array like a chessboard: (i, j) is black when $i + j$ is odd. The condition is stated in terms of the DIAGONAL and ANTIDIAGONAL neighbours of a cell, that is the four offsets

$$(-1, -1), (1, 1), (-1, 1), (1, -1),$$

and for a cell (i, j) we write $S(i, j)$ for those of them that land inside the array. The condition is that every cell has a neighbour one greater than it modulo 4, and no neighbour equal to it:

$$\exists (a, b) \in S(i, j) \quad x(a, b) \equiv x(i, j) + 1 \pmod{4}, \quad x(a, b) \neq x(i, j) \text{ for all } (a, b) \in S(i, j).$$

The entry adds that the first cell of the subarray in reading order, the cell $(0, 1)$ of the array, is 0.

Lemma 1. *Each of the four offsets changes i and j each by one, so it changes $i + j$ by 0 or ± 2 and never changes its parity. Every neighbour of a black cell is therefore black. The condition above splits into two systems that share no cell: one on the black cells and one on the white cells.*

So what the entry counts is the assignments of values to the black cells alone that satisfy the black system; the white cells never enter. That this is the entry's own reading, and not merely a convenient one, is settled by its "upper left element zero": for a black subarray the first cell in reading order is the cell $(0, 1)$ of the array, and it is that cell the clause fixes, as the published terms confirm and as Section 5 records.

3 A walk on one colour

Take the array one row at a time. In a row of even index the black cells sit at every other column starting from column 1, and in a row of odd index they sit at the columns in between; the two shapes alternate, so a state must record which of them the next row has.

All four offsets move exactly one row, so the neighbours of the black cells of a row are the black cells of the rows immediately above and below it. A row's conditions are therefore settled by three consecutive rows, and the state is the last two of them together with the parity of the next.

Appending a row completes one such window and settles the row in the middle of it. Because the condition looks one row DOWN as well as up, the last row of a finished array has not been tested when the walk stops, so the end vector is not all ones: a state is accepted exactly when its last row passes with nothing below it. Thus

$$a(n) = \iota^\top M^j \tau, \quad S = 157.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C -finite.

4 The proof

Lemma 2. *Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+3} - \sum_i c_i M^{j+3-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) = 3a(n-1) - a(n-2) + 3a(n-3)$$

for every $n > 1$.

Proof. The vector $w = q(M)\tau$ was formed by 3 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 1 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 26 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: the cells of the black colour were listed from $i + j$ directly, every assignment of values to them enumerated, and each tested by the entry's own words. No rows, no window, no state and no end vector are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.